MATH 435 Preliminary Data Analysis and Conclusions

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1 Project Question and Current Direction

Although our proposal initially focused on gathering data to investigate the relationship between food access and potential health outcomes, our preliminary analysis has shifted our focus toward a correlation analysis between health, socioeconomic, and environment factors using the Census Tracts of Lancaster County. We are now investigating the hypothesis that food environment density in Lancaster County is a weak predictor of health outcomes, and instead analyzing the impact of poverty, lack of vehicle access, high school completion status, and insurance status.

Central Question: To what extent do socioeconomic variables (education, poverty, and insurance status) and transportation barriers outweigh physical proximity in determining chronic health outcomes (Obesity, Diabetes, and CHD) in Lancaster County?

Refined Aims: While our initial proposal focused on "food deserts" as a matter of physical distance, our current direction focuses on the socioeconomic effects. We aim to identify tracts where health risks are highest not because grocery stores are far, but because residents lack the financial and physical means to access them.

Prior research in Nebraska has identified strong relationships between obesity, food insecurity, and socioeconomic disadvantage. Studies indicate that children in low-income households and those experiencing food insecurity exhibit significantly higher obesity rates, and disparities also exist across education levels, race, and access to healthy food environments [Neb15]. These findings motivate our focus on socioeconomic predictors and interaction effects rather than geographic distance alone.

2 Data Description and Processing

2.1 Data Sources and Content

Our analysis utilizes a synthesized dataset consisting of 74 neighborhoods (Census Tracts) in Lancaster County:

Census Tract: Defined as a relatively small subdivision of a county determined by population characteristics. These subdivisions fit like puzzle pieces to form a county. [U.S94]

The synthesized dataset was constructed by the merging of three primary datasets or pieces of information:

1. **Health Statistics** [CDC]: The percentage of people in each area with Obesity, Diabetes, and Coronary Heart Disease.

2. **Socioeconomic Indicators and Statistics** [Cen]: The "poverty rate" (percentage of households below the federal poverty line), the "uninsured rate", and the "percent no high school rate". These are our primary independent variables.

3. **Spatial and Access Variables/Statistics** [Boe]: Includes the physical distance to the grocery stores (in km), the percentage of households without a vehicle, and store density (points per square km).

The preliminary analysis focuses on 74 census tracts within Lancaster County. These tracts were selected because they contain complete data across all variables. Tracts with missing values were

excluded to ensure consistency in regression and correlation analysis. This subset provides a balanced representation of both urban and rural areas within the county.

2.2 Data Relevance

The most relevant features for our study are socioeconomic variables and mobility-based access measures. Rather than examining distance to grocery stores alone, we consider both distance and transportation limitations as separate predictors.

Preliminary correlation analysis indicates that percent without a high school diploma exhibits the strongest relationship with obesity ($r \approx 0.75$). Food access variables such as distance to grocery stores and percent without a vehicle show weaker relationships, suggesting that socioeconomic conditions may play a larger role in determining health outcomes.

2.3 Data Processing Procedures

To make sure our data was clean and organized, we wrote a script to pick out only the most important and relevant pieces from the datasets for our study. We used the following code to narrow down our list and format the features with tract as a key:

Listing 1: Selecting Key Variables for Analysis

```
cols_to_keep = [
    'tract',
    'DIABETES_CrudePrev',
    'OBESITY_CrudePrev',
    'CHD_CrudePrev',
    'percent_uninsured',
    'fast_food_density',
    'grocery_density',
    'fast_food_ratio',
    'no_grocery',
    'dist_to_fast_food_km',
    'dist_to_grocery_km',
    'percent_no_car',
    'access_constraint',
    'poverty_rate',
    'median_income',
    'percent_no_hs'
]
```

Final Dataset Preparation

After selecting our variables, we merged and saved our final working dataset using the logic below:

Listing 2: Filtering and Exporting the Master File

```
# Filter for the columns defined above
df_final = df_full[cols_to_keep]

# Export the master file for the team
df_final.to_csv("Lancaster_Full_Dataset.csv", index=False)
```

The Cleaned Data Result

The resulting file is a simple table where every row is a neighborhood. Because we formatted the IDs as text, we can easily map this data. This organization makes it possible for us to see, for example, if neighborhoods with the lowest income also have the highest rates of heart disease.

Why We Used This Specific Data

We are using these 74 neighborhoods or "Tracts" because they represent the entire "story" of the county. By analyzing the difference between the rural and urban parts of Lancaster County, we can test if our theories, such as if having a car is more important than living close to a store, holds true for all different types of residents and factors being investigated.

2.4 Cleaned Data Tables

The following three tables display the first five tracts of Lancaster County and the relationship with the data retrieved and cleaned from the primary datasets:

Table 1: Preliminary Health Data Snapshot (Lancaster County)

Neighborhood (Tract)	**Diabetes (%)**	**Obesity (%)**	**CHD (%)**
1.00	12.8	40.1	7.6
2.01	11.0	35.9	7.0
2.02	9.1	37.3	4.9
3.00	10.2	37.8	5.5
4.00	11.8	41.1	5.7

Table 2: Socioeconomic and Access Snapshot (Lancaster County)

Neighborhood (Tract)	**Poverty Rate** (%)	**No Car (%)**	**Grocery Dist (km)**
1.00	12.8	13.8	5.56
2.01	16.4	10.8	4.63
2.02	23.8	9.1	3.44
3.00	19.9	7.8	2.72
4.00	30.9	4.5	0.73

Table 3: County-Wide Averages and Ranges (N=74 Neighborhoods)

Variable	**Average (Mean)**	**Lowest (Min)**	**Highest (Max)**
Obesity Rate (%)	34.9	30.7	42.4
Poverty Rate (%)	13.4	1.3	63.7
No Car (%)	6.6	0.0	33.5
Distance to Grocery (km)	5.0	0.2	22.0

3 Assumptions and Setup

To make our mathematical model work correctly, we have made a few key assumptions. These assumptions help us focus on the most important parts of food access in Lincoln without getting distracted by every single small detail.

1. **Transportation is the Master Key**

We assume that in a city like Lincoln, owning a vehicle is the most important factor for gaining access to healthy food sources.

- **The logic:** If you have a car, living 5 miles from a grocery store is just a 10-minute drive. But if you do not own a car, that same distance might require an hour-long bus trip while carrying heavy bags. Therefore, we must assume that distance only matters if you don't own a car.

2. **Grocery Store vs. Fast Food**

We assume that "Grocery Stores" (such as Hy-Vee, Russ's, or Super Saver) are the primary sources of healthy food, while "Fast Food" locations are primarily sources of high-calorie, lower-nutrition food.

- **The Logic:** While some grocery stores sell junk food and some fast-food places sell salads, a neighborhood with more grocery stores and fewer fast-food heatmaps should statistically have better health outcomes.

3. **Data Consistency**

We assume that the information from our different sources (CDC, Census, and USDA) is consistent enough to compare.

- **The Logic:** Even though some data might be from 2023 and some from 2024, neighborhood characteristics, such as where the stores are and which areas are low-income, usually stay the same for several years. We assume these small timing differences won't change our overall results.

4. **The "Urban-Rural" Mix**

We assume that Lancaster County is a good "test site" for the rest of the region.

- **The Logic:** Because Lincoln has a mix of a very crowded city center and very open rural farmland, we assume that if our model works here, it could eventually be used to help other cities in Nebraska or other Midwest States.

3.1 Setup and Definitions

For our study, we have defined a "High Vulnerability Area" as any neighborhood that meets two criteria:

- 1. It is in the top 20 percent for poverty.
- 2. It has a high Access Constraint (meaning stores are far and many people do not have cars).

Food Insecurity: Defined as limited or uncertain availability of nutritionally adequate and safe foods or limited or uncertain ability to acquire acceptable foods in socially acceptable ways.

By setting these ground rules, we ensure that our final analysis focuses on the people who are most likely to be struggling with food access, and have a greater probability for being at risk for the chosen health diseases.

3.2 Statistical Model Assumptions

We assume that the linear regression model satisfies standard assumptions:

- Linearity between predictors and response
- Independence between census tracts
- Constant variance of residuals
- Approximately normally distributed errors

These assumptions allow the least squares estimator to provide unbiased coefficient estimates.

4 Method and Preliminary Analysis

4.1 Variable Definitions

Let $i = 1, 2, \ldots, n$ index census tracts in Lancaster County. We define the response variable

$$Y_i = \text{Obesity prevalence in tract } i$$

and explanatory variables

$$X_{1i} = \text{Percent of population without a high school diploma}$$

$$X_{2i} = \text{Fast food density (locations per km}^2\text{)}$$

$$X_{3i} = \text{Distance to nearest grocery store (km)}$$

These variables were selected using preliminary correlation analysis to identify strong predictors while avoiding multicollinearity among socioeconomic indicators.

4.2 Correlation Matrix

To examine relationships, we computed the correlation matrix

$$R = (r_{jk})$$

where

$$r_{jk} = \frac{\sum_{i=1}^{n}(X_{ji} - \bar{X}_j)(X_{ki} - \bar{X}_k)}{\sqrt{\sum_{i=1}^{n}(X_{ji} - \bar{X}_j)^2}\sqrt{\sum_{i=1}^{n}(X_{ki} - \bar{X}_k)^2}}$$

This matrix quantifies linear relationships between variables.[Tea]

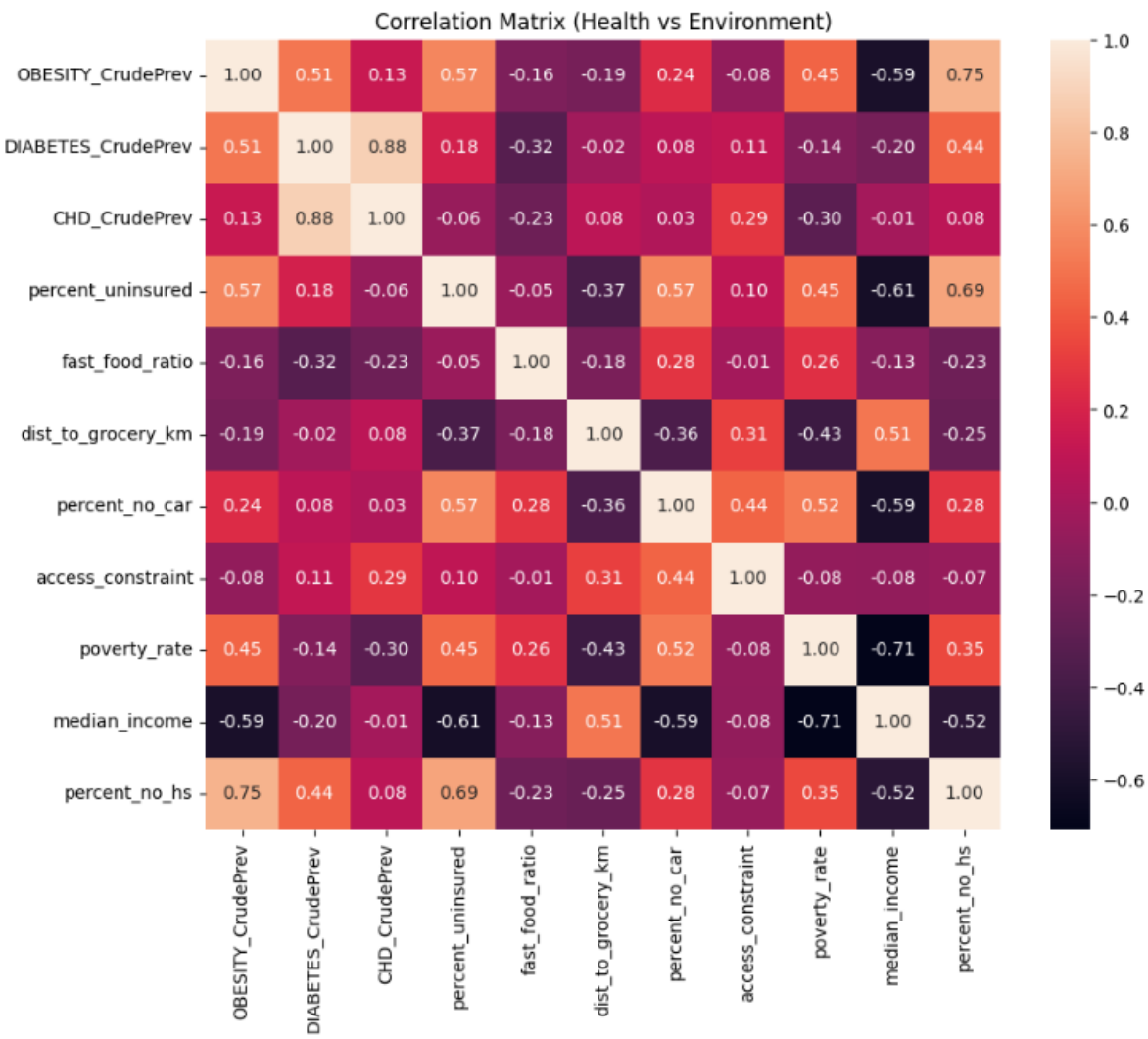

Figure 1: Correlation matrix for health, socioeconomic, and food access variables. Strong correlations are observed between obesity and socioeconomic indicators, particularly percent without high school diploma.

Strong correlations were observed between obesity and socioeconomic indicators, particularly percent without a high school diploma. Food environment variables such as fast food density and distance to grocery stores exhibited weaker correlations.

Because several socioeconomic variables were highly correlated with one another, percent without a high school diploma was selected as the primary socioeconomic predictor.

4.3 Linear Regression Model

We model obesity prevalence using a multiple linear regression model

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + \varepsilon_i$$

where

$$\varepsilon_i \sim N(0, \sigma^2)$$

represents unexplained variation across census tracts.

Here, X_{1i} represents percent without high school diploma, X_{2i} represents fast food density, and X_{3i} represents distance to the nearest grocery store in kilometers.

These variables were selected based on both correlation analysis and conceptual relevance. Percent without a high school diploma exhibited the strongest correlation with obesity ($r \approx 0.75$), making it the primary socioeconomic predictor. Fast food density was included to capture differences in the local food environment, as areas with higher concentrations of fast food outlets may provide greater access to calorie-dense foods. Distance to grocery stores was included to measure geographic access to healthier food options. Together, these variables allow the model to compare socioeconomic disadvantage with physical food accessibility.

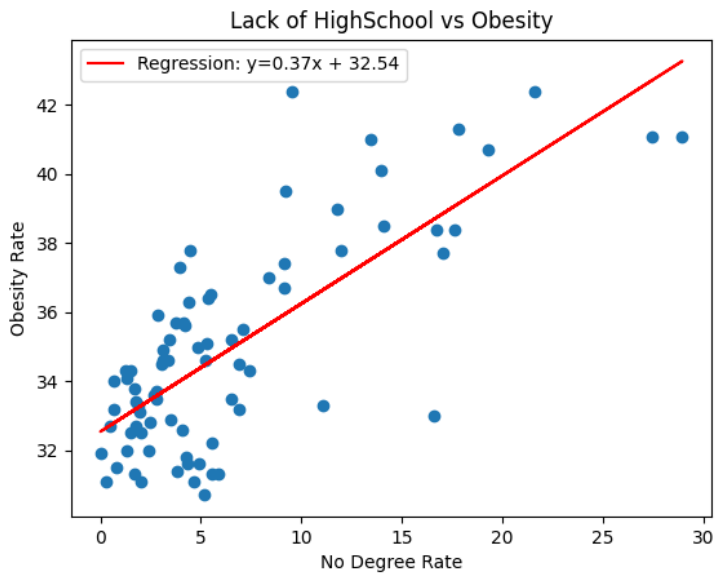

Figure 2: Obesity prevalence versus percent without high school diploma. A strong positive linear trend is observed.

4.4 Matrix Formulation

The regression model can be written in matrix form

$$\mathbf{Y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

where

$$\mathbf{Y} = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}$$

$$X = \begin{bmatrix} 1 & X_{11} & X_{21} & X_{31} \\ 1 & X_{12} & X_{22} & X_{32} \\ \vdots & \vdots & \vdots & \vdots \\ 1 & X_{1n} & X_{2n} & X_{3n} \end{bmatrix}$$

$$\boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix}$$

4.5 Least Squares Estimation

The coefficients are estimated using ordinary least squares

$$\hat{\boldsymbol{\beta}} = (X^T X)^{-1} X^T \mathbf{Y}$$

which minimizes the residual sum of squares

$$RSS = \sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2$$

where

$$\hat{Y}_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i}$$

4.6 K-Fold Cross Validation

To evaluate model stability, we implemented K-fold cross validation with $K = 5$. The dataset was partitioned into disjoint subsets

$$D = D_1 \cup D_2 \cup \cdots \cup D_K$$

For each fold k, the model was trained on

$$D_{-k} = D \setminus D_k$$

and evaluated on D_k.
The cross-validated coefficient of determination is

$$R^2_{CV} = \frac{1}{K} \sum_{k=1}^{K} R_k^2$$

This procedure reduces dependence on a single train-test split.

4.7 Algorithmic Implementation

The method was applied as follows:

1. Clean and merge tract-level datasets
2. Compute correlation matrix
3. Select predictors with strongest correlations
4. Fit linear regression model
5. Evaluate model using R^2
6. Perform 5-fold cross validation
7. Visualize spatial distribution of obesity

4.8 Model Interpretation

The regression model achieved

$$R^2 \approx 0.49$$

indicating that approximately 49% of variation in obesity prevalence is explained by the selected predictors. This is visualized in the following plot.

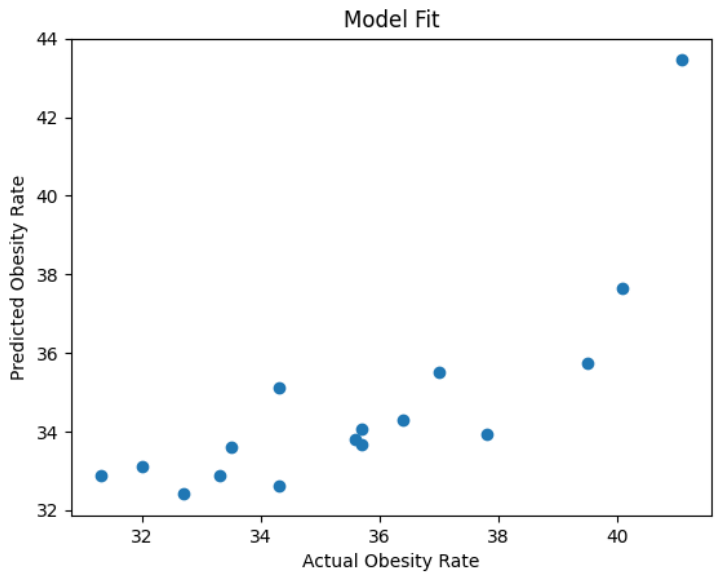

Figure 3: Predicted versus observed obesity prevalence. The positive linear trend indicates moderate model fit.

Cross validation produced

$$R^2_{CV} \approx 0.36$$

indicating moderate predictive power and variability across census tracts.

These results suggest that socioeconomic factors, particularly education level, are stronger predictors of obesity than physical proximity to food outlets. We also tested models using poverty rate and uninsured rate as predictors. These models produced lower R^2 values than percent without high school diploma, confirming that education level is the strongest single predictor.

While the regression model quantifies statistical relationships between variables, it does not capture the geographic structure of health outcomes. To better understand spatial variation across Lancaster County, we visualize obesity prevalence by census tract.

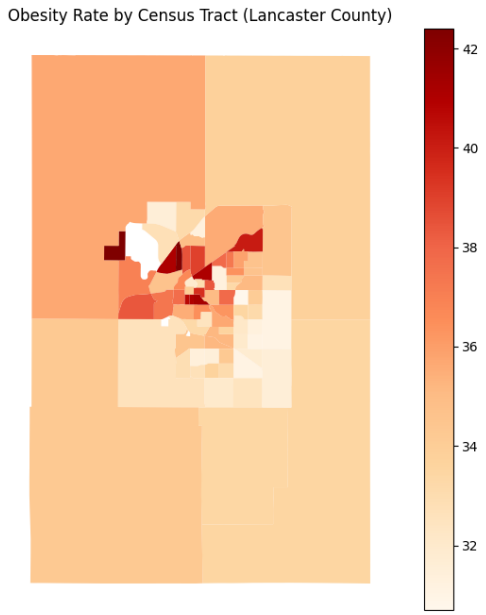

Figure 4: Spatial distribution of obesity prevalence across Lancaster County census tracts.

The spatial distribution shows clear clustering of higher obesity prevalence in specific regions of the county, with lower rates appearing in the south of Lincoln. These clusters correspond more closely with areas of socioeconomic disadvantage than with geographic distance to food outlets. This supports the regression results indicating that education and economic variables are stronger predictors of health outcomes than food access alone.

5 Preliminary Conclusions

5.1 Summary of Findings

Our preliminary analysis suggests that geographic proximity to grocery stores is a weaker predictor of obesity than socioeconomic background.

These findings are consistent with prior research in Nebraska showing that obesity disparities are strongly associated with poverty, education, and food insecurity rather than geographic access alone [Neb15].

The Power of Education:

Our regression model showed that the "No High School Diploma" rate was the strongest predictor of obesity. In simple terms, for every increase in the number of people without a degree in a neighborhood, we see a predictable rise in obesity, regardless of how many grocery stores are nearby.

- **The Distance Problem:** While many people talk about "Food Deserts" as a problem of physical distance, our correlation matrix showed that distance to a grocery store had a very weak relationship with health outcomes in Lancaster County. This confirms our earlier assumption that having a car or the money to travel makes physical distance a lesser factor for many residents.

- **Regression analysis:** Our model explains about 49% of the differences in health between neighborhoods. While that is a strong start, it also tells us that there is another 51 percent of "missing information" we have not captured yet (likely the cost of food, cultural habits, or the quality of public transit).

Significance of the Model:

The map in the previous section proves that obesity is not scattered randomly; it "clusters" in specific parts of Lincoln. Because the model's predictions align with these clusters ($R^2 \approx 0.49$), we know we are on the right track. We have successfully moved the project away from a simple "map of stores" to a "map of vulnerability."

5.2 Next Steps to Strengthen the Analysis

To move this project toward our final paper, our group will take the following steps:

1. **Add the "No Car" Factor:** Since our group member found that distance alone is weak, we will add our Access Constraint variable (Distance × No Car) into the regression model. We want to see if this "interaction" explains more than 49% of the variation.

2. **Outlier Investigation:** We noticed some neighborhoods on the map that "defy the rules"—areas with high poverty but surprisingly low obesity. We want to look closer at these specific tracts (Points of Interest) to see if they are near community gardens, high-frequency bus stops, or health centers.

3. **Refine the Regression:** We will check for "Residuals" (the errors in our model). If the model consistently misses the mark in the same parts of town, it means there is a geographic factor we haven't accounted for yet.

These findings directly address the central question by showing that socioeconomic variables outweigh geographic food access in predicting health outcomes.

References

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